DUTCH-BANGLA BANK

WRITTEN MATH

Previous Questions with Step-by-Step Solutions

Written Math by

RASHED MENON

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For Bank Job Written Preparation

Job Study With Rashed

2024 - Sales Manager (Written)

Exam Date: 18.09.2024 | Exam Taker: IBA

Question 1

The sum of two digits of a number is 16. If the digits of this number are interchanged, then the new number is 18 greater than the original number. What is the original two-digit number?

Solution:

Let the tens digit = x and the units digit = y.

x + y = 16 .....(1)

Original number = 10x + y

Interchanged number = 10y + x

10y + x = 10x + y + 18

9y - 9x = 18

y - x = 2 .....(2)

Adding (1) and (2): 2y = 18, so y = 9

Therefore, x = 16 - 9 = 7

Original number = 79

Answer: 79

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Question 2

Dolar can do a piece of work in 4 hours; Zoha and Khadem together can do it in 3 hours, while Dolar and Khadem together can do it in 2 hours. How long will Zoha alone take to do it?

Solution:

Dolar's 1-hour work = 1/4

Zoha + Khadem's 1-hour work = 1/3

Dolar + Khadem's 1-hour work = 1/2

Khadem's 1-hour work = 1/2 - 1/4 = 1/4

Zoha's 1-hour work = 1/3 - 1/4 = 1/12

Therefore, Zoha alone will take 12 hours.

Answer: 12 Hours

Note: The source mentions 12 Days, but the correct unit from the given data is Hours.

Question 3

The percentage profit selling a product for Tk. 1,920 is equal to the percentage loss incurred by selling the same product for Tk. 1,280. What price should the product be sold to make 25% profit?

Solution:

When profit percentage and loss percentage are equal:

Cost Price = (Higher Selling Price + Lower Selling Price) / 2

= (1920 + 1280) / 2

= 3200 / 2 = Tk. 1,600

For 25% profit:

Selling Price = 1600 x 125/100 = Tk. 2,000

Answer: Tk. 2,000

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2023 - Trainee Officer (Written)

Exam Date: 11.11.2023 | Exam Taker: IBA

Question 1

Of the 3 machines Company X uses for production, Machine A can produce 30 units in 5 hours, Machine B can produce 6 units in 12 hours, and Machine C can produce 33 units in 12 hours. Of the 30 hours the company uses the facility, the committee decided they would operate Machine B and C for 2/5 of the total time and Machine A and B to produce 39 units. Given the committee decided to operate Machine A and C for the rest of the time, what are the total units produced by the company?

Solution:

Machine A rate = 30/5 = 6 units/hour
Machine B rate = 6/12 = 0.5 unit/hour
Machine C rate = 33/12 = 2.75 units/hour
Total available time = 30 hours
B and C operate for 30 x 2/5 = 12 hours
B + C rate = 0.5 + 2.75 = 3.25 units/hour
Production by B + C = 3.25 x 12 = 39 units
A + B rate = 6 + 0.5 = 6.5 units/hour
Time for A + B to produce 39 units = 39/6.5 = 6 hours
Time used = 12 + 6 = 18 hours; remaining time = 12 hours
A + C rate = 6 + 2.75 = 8.75 units/hour
Production by A + C = 8.75 x 12 = 105 units
Total production = 39 + 39 + 105 = 183 units

Answer: 183 Units

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Question 2

Dhaka Express and Khulna Express trains left from two stations towards Khulna station and Dhaka station respectively. Three hours after they met, the trains were 675 miles apart. The Dhaka Express arrived at its destination 16 hours after their meeting and the Khulna Express arrived at its destination 25 hours after their meeting. How many hours did Dhaka Express take for the whole trip?

Solution:

For two trains starting simultaneously and meeting once:

Time before meeting = sqrt(t1 x t2)

Here, t1 = 16 hours and t2 = 25 hours

Time before meeting = sqrt(16 x 25) = sqrt(400) = 20 hours

Dhaka Express took 16 hours after meeting.

Total journey time = 20 + 16 = 36 hours

Answer: 36 Hours

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Question 3

Babu purchased 6 identical maple wood furniture and 7 identical birch wood furniture for his office. The regular price of each maple wood furniture was thrice the regular price of each birch wood furniture. However, Babu obtained a 20% discount off the regular price of each maple wood furniture. If the regular price of each birch wood furniture was Tk. 500, what was the total price of all 13 furniture?

Solution:

Birch furniture price = Tk. 500 each

Regular maple price = 500 x 3 = Tk. 1,500

After 20% discount, maple price = 1500 x 80/100 = Tk. 1,200

Price of 6 maple furniture = 1200 x 6 = Tk. 7,200

Price of 7 birch furniture = 500 x 7 = Tk. 3,500

Total price = 7200 + 3500 = Tk. 10,700

Answer: Tk. 10,700

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2023 - Trainee Officer (Written)

Exam Date: 03.02.2023 | Exam Taker: IBA

Question 1

On selling an article for Rs. 400, a person loses 20% of its selling price. At what price should he sell it to gain 15% on its cost price?

Solution:

Solution according to the provided answer:

If Tk. 400 represents 80% of the Cost Price:

Cost Price = 400 x 100/80 = Tk. 500

For 15% profit:

Selling Price = 500 x 115/100 = Tk. 575

Answer: Tk. 575

Note: If the wording "loses 20% of its selling price" is taken literally, the answer would be Tk. 552. Tk. 575 is correct if the loss is 20% of cost price.

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Question 2

A man deposited Tk. 10,000 for 2 years and Tk. 5,000 for 3 years at the same rate of simple interest. If he received a total interest of Tk. 5,250, what is the rate of interest per annum?

Solution:

Simple Interest = PRT/100

Interest on Tk. 10,000 for 2 years = (10000 x R x 2)/100 = 200R

Interest on Tk. 5,000 for 3 years = (5000 x R x 3)/100 = 150R

Total interest: 200R + 150R = 5250

350R = 5250

R = 5250/350 = 15

Rate of interest = 15% per annum

Answer: 15% per annum

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2022 - Management Trainee Officer (Written)

Exam Date: 01.04.2022 | Exam Taker: IBA

Question 1

A person has two stamps, one is US Stamp and another is European Stamp. US stamps are sold at 30% profit and European stamps are sold at 20% loss. The original purchase prices of US Stamp and European Stamp are Tk. 18,000 and Tk. 20,000 respectively. What is the total revenue earned from selling the stamps?

Solution:

Purchase price of US Stamp = Tk. 18,000

Selling price of US Stamp = 18,000 x 130/100 = Tk. 23,400

Purchase price of European Stamp = Tk. 20,000

Selling price of European Stamp = 20,000 x 80/100 = Tk. 16,000

Total revenue = 23,400 + 16,000 = Tk. 39,400

Answer: Tk. 39,400

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Question 2

Machine A takes 2 hours to make a car, while Machine B takes 3 hours to make the same car. After working together for 1 hour, Machine B breaks down. How much additional time will Machine A take to complete the car?

Solution:

Machine A's 1-hour work = 1/2

Machine B's 1-hour work = 1/3

Together in 1 hour = 1/2 + 1/3 = 5/6

Remaining work = 1 - 5/6 = 1/6

Time required by A = (1/6) / (1/2) = 1/3 hour

= 1/3 x 60 = 20 minutes

Answer: 20 Minutes

Note: The source gives 30 minutes, but the mathematically correct answer from the stated data is 20 minutes.

Question 3

Joynol and Saba run around a circle. Joynol runs at a constant rate of 1,200 feet per minute and Saba runs at a constant rate of 800 feet per minute. If Joynol is 80 feet behind Saba, how many seconds will it take for Joynol to overtake Saba and become 40 feet ahead of Saba?

Solution:

Joynol's speed = 1,200 feet/minute

Saba's speed = 800 feet/minute

Relative speed = 1,200 - 800 = 400 feet/minute

Required gain = 80 + 40 = 120 feet

Time = 120/400 minute = 3/10 minute

= 3/10 x 60 = 18 seconds

Answer: 18 Seconds

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2019 - Probationary Officer (Written)

Exam Date: 15.11.2019 | Exam Taker: IBA, DU

Question 1

In a college of 100 students, 60% took Finance, 45% took Marketing, and 45 students took Operations Management. Ten students took all three courses, and ten students took none of these courses. How many students took exactly two of these three courses?

Solution:

Finance = 60, Marketing = 45, Operations Management = 45

Students taking at least one = 100 - 10 = 90

Let x = exactly one course and y = exactly two courses

x + y + 10 = 90, so x + y = 80 .....(1)

Total course registrations = 60 + 45 + 45 = 150

x + 2y + 3(10) = 150

x + 2y = 120 .....(2)

Subtracting (1) from (2): y = 40

Answer: 40 Students

Question 2

During a tree plantation drive, a rectangular plot of land measuring 90 feet by 66 feet is surrounded by a boundary wall. Trees will be planted, leaving a 5-foot gap from the wall on all sides. The distance between adjacent trees in both rows and columns will be 4 feet. What is the maximum number of trees that can be planted?

Solution:

Usable length = 90 - 5 - 5 = 80 feet

Usable width = 66 - 5 - 5 = 56 feet
Trees along length = 80/4 + 1 = 21
Trees along width = 56/4 + 1 = 15
Maximum number of trees = 21 x 15 = 315

Answer: 315 Trees

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2017 - Probationary Officer (Written)

Exam Date: 18.08.2017 | Exam Taker: IBA, DU

Question (a)

Nipu sold 100 pens, of which 50 were red and 50 were black, at Tk. 48 per pen. He made a profit of 20% on the black pens and a loss of 20% on the red pens. What was the net gain or net loss on this sale?

Solution:

Selling price of each pen = Tk. 48

Black pen cost price = 48 x 100/120 = Tk. 40

Cost of 50 black pens = 40 x 50 = Tk. 2,000

Red pen cost price = 48 x 100/80 = Tk. 60

Cost of 50 red pens = 60 x 50 = Tk. 3,000

Total Cost Price = Tk. 5,000

Total Selling Price = 100 x 48 = Tk. 4,800

Net Loss = 5,000 - 4,800 = Tk. 200

Answer: Net Loss = Tk. 200

Question (b)

A trader purchases blue paint and yellow paint at Tk. 110 and Tk. 100 per liter respectively. He mixes blue and yellow paint at a ratio of 1:2 to make green paint. He sells the green paint at Tk. 120 per liter. If the trader makes a profit of Tk. 2,000, how much yellow paint did he use in the mixture?

Solution:

Blue : Yellow = 1 : 2

Take 1 liter blue and 2 liters yellow.

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Cost of 3 liters mixture = 110 + 2(100) = Tk. 310

Selling price of 3 liters = 3 x 120 = Tk. 360

Profit per 3-liter batch = 360 - 310 = Tk. 50

Number of batches for Tk. 2,000 profit = 2000/50 = 40

Yellow paint = 40 x 2 = 80 liters

Assuming 1 liter approximately equals 1 kg, yellow paint used = 80 kg

Answer: 80 Kg

Note: The source appears to show 1:21. To obtain the provided answer of 80 kg, the intended ratio must be 1:2.

Question (c)

In a class of 25 students, 10 have less than 6 marbles, 10 have more than 7 marbles and 4 have more than 8 marbles. How many students have more than 5 marbles but less than 9 marbles?

Solution:

Total students = 25

Students with less than 6 marbles = 10

Students with more than 8 marbles = 4

Required students have 6, 7, or 8 marbles.

Required number = 25 - 10 - 4 = 11

Answer: 11 Students

Question (d)

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A ship with a faulty engine sailed for only 7 hours over a period of 2 days and covered a distance of 59 km. Its average speed on the first day was 5 km per hour slower than that on the second day. It sailed 3 hours longer on the first day than on the second day. What was its average speed on the second day?

Solution:

Let second-day sailing time = x hours.

First-day sailing time = x + 3 hours.

$x + (x + 3) = 7$

$2x = 4$, so $x = 2$

Second-day time = 2 hours; first-day time = 5 hours

Let second-day speed = v km/h.

First-day speed = v - 5 km/h.

$5(v - 5) + 2v = 59$

$7v = 84$

$v = 12$ km/h

Answer: 12 km/h

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2017 - Assistant Officer (Written)

Exam Date: 18.03.2017 | Exam Taker: IBA, DU

Question (a)

A picnic was arranged by m students. The total cost of the picnic was estimated to be y taka. Unfortunately, z students withdrew their names from the picnic. How many more taka would each of the remaining students have to pay?

Solution:

Initial contribution per student = y/m

Remaining students = m - z

New contribution per student = y/(m - z)

Additional amount = y/(m - z) - y/m

= [ym - y(m - z)] / [m(m - z)]

= yz / [m(m - z)]

Answer: yz / [m(m - z)] Taka

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Question (b)

A number consists of 3 digits whose sum is 10. The middle digit is equal to the sum of the other two digits, and the number will be increased by 99 if the first and last digits are reversed. What is the number?

Solution:

Let the digits be a, b, c.

a + b + c = 10 .....(1)

b = a + c .....(2)

Substitute into (1): 2a + 2c = 10, so a + c = 5 and b = 5.

Original number = 100a + 10b + c
Reversed number = 100c + 10b + a
Increase by 99 gives: 99c - 99a = 99
c - a = 1 .....(3)
Using a + c = 5 and c - a = 1:
2c = 6, so c = 3 and a = 2
Therefore, b = 5 and the number is 253.

Answer: 253

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